

Chap 4

1. What is project?

- A project is a (temporary) ~~unique~~ sequence of unique,

complex and connected activities that have

one goal or purpose and that must be completed by a specific time, within budget and according to specification.

- Characteristics of a project:

1. A clear start and end date:

Example: There are projects that last several years but projects cannot go on forever.

• ~~It~~ It needs to have a clear beginning, a definite end and an overview of

~~what~~ what happens in between. (I)

(II) clear objectives and

2. A project creates something new

- Every project is unique, producing something that did not previously exist.

3. A project has boundaries:

- A project operates within certain constraints of time, money, quality and functionality.

4. A project is not business as usual:

- Project is not business, its part of a business.

Q: What is the major cause of project failure?

• Project may fail due to several reasons,

(i) Failure to establish upper-management commitment to the project.

(ii) Lack of organization's commitment to the system development methodology.

(iii) Poor expectations management.

(a) Scope creep

(b) Feature u

(IV) Premature commitment to a fixed budget, and schedule.

(V) Overoptimism

(VI) Inadequate people management skills.

(VII) Failure to adapt to business change

(VIII) Insufficient resources

(IX) Failure to "manage to the plan".

Ultimately the major cause of project failure is that, most project managers were not educated or trained to be project managers.

• Good system analysts don't automatically perform well as project managers.

• To be a good project manager, one should be

educated and skilled in the "art" of project management".

Q What is the difference between scope creep and feature creep?

Scope

- Is the unexpected growth of user expectations and business requirements for an information system as the project progresses.

- Applies to projects

- scope creep and feature creep are not correlated.

Feature

- Is the uncontrolled addition of technical features to a system under development without regard to schedule or budget.

- Applies to products.

- scope creep doesn't cause feature creep.

Q] What are the five main categories of competencies that a project manager should have?

Answer:

1/ Business Achievement competencies.

• It includes competencies like

(i) Business awareness

(ii) Business partnership orientation

(iii) Commitment to quality

2/ Problem solving competencies.

• Includes .

(i) Initiative

- demonstrates creativity, calculated risks, and persistence necessary to get the job done.

(ii) Information gathering.

(iii) Analytical thinking.

(iv) conceptual thinking.

3// Influence competences

- (i) Interpersonal awareness.
- (ii) Organizational "
- (iii) Anticipation of impact.
- (iv) Resourceful use of influence.

4// People Management competencies

- (i) motivating others
- (ii) communication skills
- (iii) developing others
- (iv) monitoring and controlling.

5// Self Management competencies

- (i) self confidence
- (ii) stress management
- (iii) concern for credibility
- (iv) Flexibility.

Q Why are business achievement competencies important?

• ~~Proj~~ A project is a part of a business. If a project manager doesn't have business achievement competencies then the project will not be able to achieve the business goals.

• There are two basic premises of project management competencies:

(I) Individuals can not manage a process they have never used.

(II) Managers must have an understanding of the business and culture that provides a context for the project.

• That's why business achievement competencies includes some other competencies including -

① Business awareness:

- Ties every systems project to the mission, visions and goals of the organization.

② Business partner orientation:

- keeps managers and users involved throughout a systems project.

③ Commitment to quality:

- Ensures that every systems project contributes to the quality expectation of the organization as a whole.

Q] What are the basic project management functions.

I Scoping:

(i) Scope defines the boundaries of a project.

II Planning:

(i) Planning identifies the tasks required to complete the project.

(ii) Planning is based on the manager's understanding of the project scope and methodology.

III Estimating:

• Each task that is needed to complete the project must be estimated

- ~~what~~ Time

- people

- skills

- sequence of tasks.

- cost.

(iv) Scheduling:

- With the understanding of the required tasks, task duration and task prerequisites scheduling must be done.

(v) Organizing:

- project members should be aware of individual roles, responsibilities and role relationship

(vi) Directing:

- Every project manager should demonstrate people management skills to direct team activities.

(vii) Controlling:

- It is the most important and difficult task.

VIII) cloning:

- A good project manager always assesses successes and failures at the conclusion of a project.

All these functions are dependent on ongoing interpersonal communication among the project manager, team and other managers.

Q What are PERT and Gantt charts? How to decide which one to use?

PERT Chart:

- PERT chart is a acronym for (Program Evaluation and Review technique)
- It is a graphical Network Model used to depict the interdependencies between a project task.

• Gantt chart

• a bar chart used to depict project task against a calendar.

• Each bar represents a named project task.

• The horizontal line/axis is a calendar time line

• Gantt charts offer the advantage of clearly showing overlapping tasks.

• bars can be shaded to indicate percentage completion and project progress.

PERT Chart!

Task	
Scheduled Start	Scheduled Finish
Actual Start	Actual Finish

Task	
S.S	S.F
A.S	A.F

Inter task Dependency

Scope Definition	
4-24-22	N/A
4-24-22	N/A

5-3-21

Problem Analysis	
24/5/22	24/6/22
24/5/22	24/6/22

Requirement Analysis	
25/5/22	26/5/22
25/5/22	26/5/22

5-12-21
5-12-21

5-12-21

Physical Design	
7-3-21	9-25-21
7-5-21	10-9-21

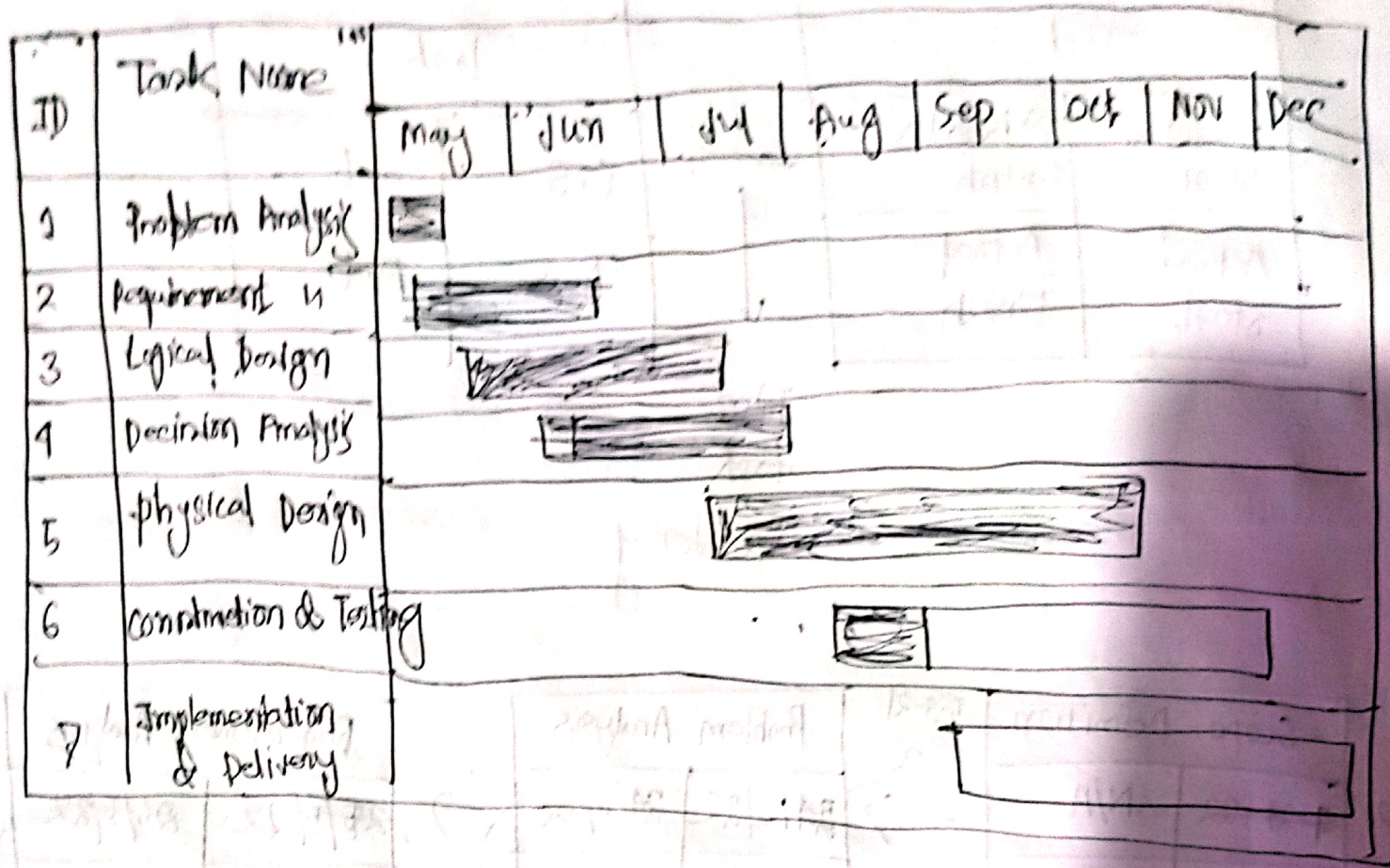
Decision Analysis	
6-13-21	7-30-21
6-13-21	8-3-21

Logical Design	
5-28-21	7-15-21
5-30-21	7-18-21

Construction/Testing	
7-19-21	11-13-21
7-20-21	(in progress)

Implementation and Delivery	
9-10-21	12-14-21
TBD	TBD

Gantt chart!



complete



Incomplete

18-8-81	18-8-81
18-8-81	18-8-81
18-8-81	18-8-81

18-8-81	18-8-81
18-8-81	18-8-81
18-8-81	18-8-81

18-8-81	18-8-81
18-8-81	18-8-81
18-8-81	18-8-81

Q. What are the eight major activities in the project management life cycle?

1. Negotiate Scope:

- scope defines the boundaries of a project.

- Five basic questions that influence the negotiation of project scope:

- product

- quality

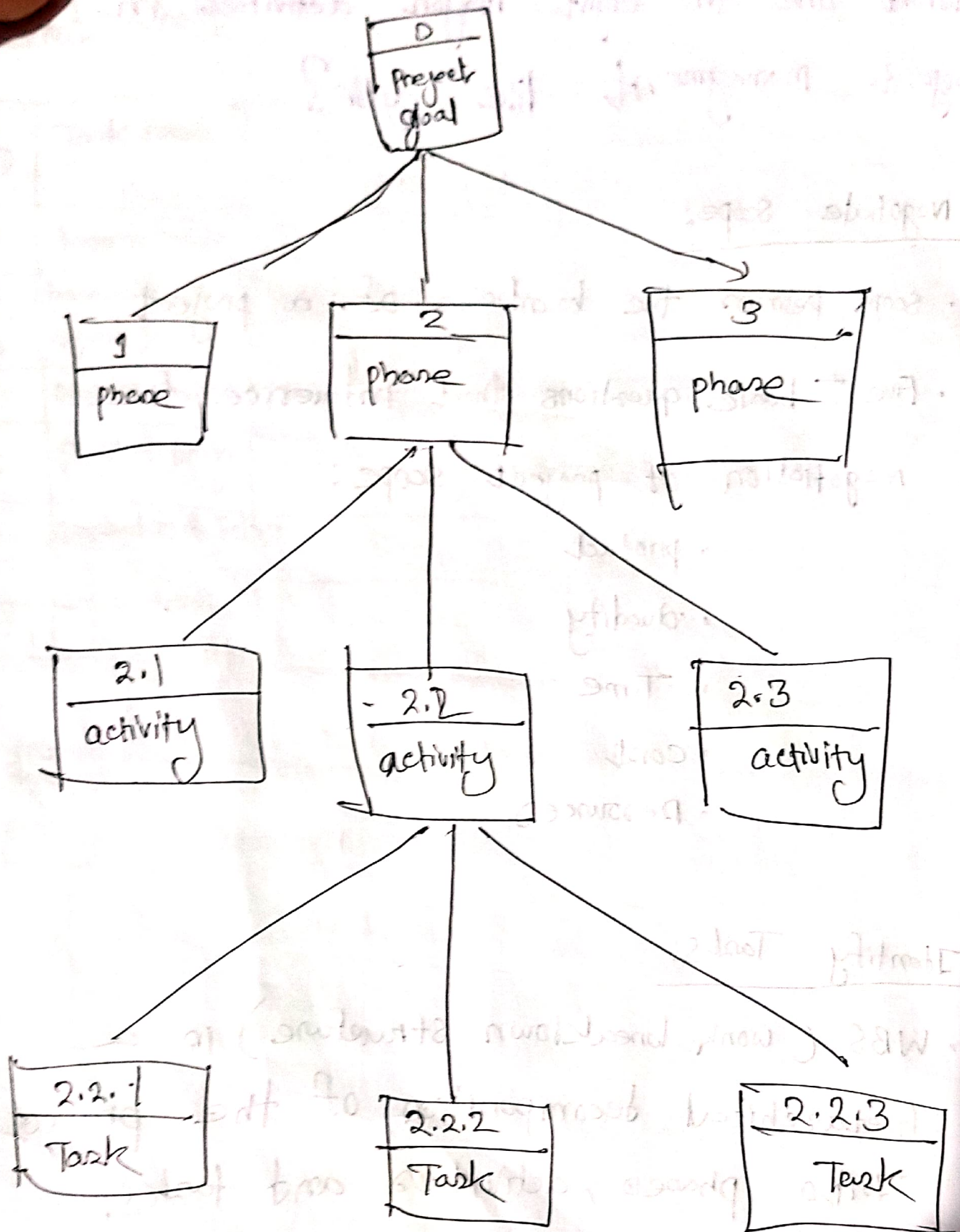
- Time

- Cost

- Resources

2. Identify Tasks:

- WBS (work breakdown structure) is a hierarchical decomposition of the project into phases, activities and tasks.



③ Estimate Task Duration:

Includes

→ optimistic Duration

- Minimum amount of time needed to complete the task.

→ pessimistic Duration

- The estimated maximum amount of time needed to complete a task.

→ Expected Duration

- Estimated amount of time required to complete a task.

→ Most likely Duration

- An Estimated amount of time required to complete a task, based on a weighted average of optimistic, pessimistic and expected duration.

$$D = \frac{(3 \times OD) + (4 \times ED) + (1 \times PD)}{C}$$

④ Specify Inter-task dependencies

• Four Types

• Finish to Start

— finish of one task triggers the start of another task.

• Start to Start

— The start of one task triggers the start of another task.

• Finish to Finish

— Two tasks must finish at the same time.

• Start to Finish

— start of one task signifies the finish of another task.

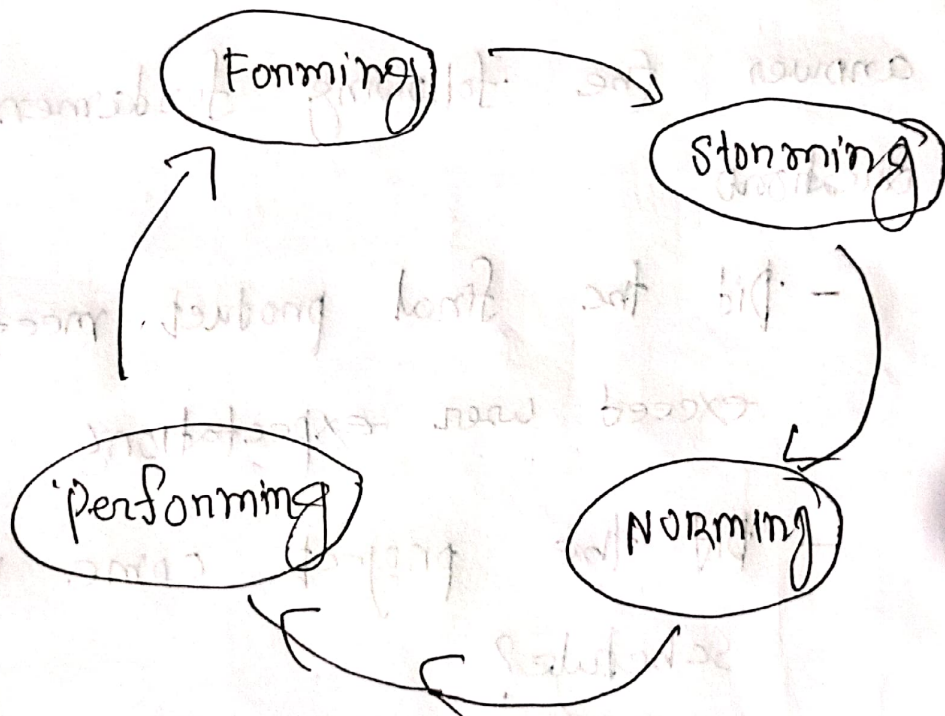
⑤ Assign Resources

• Resources includes

- people
- Services
- Facilities and equipment
- Supplies and materials
- Money

⑥ Direct the team effort!

- stages of team ~~pro~~ maturity



7. Monitor and control progress

Included tasks such as

- (i) Progress reporting
- (ii) Change management
- (iii) Expectations management
- (iv) Schedule Adjustments
— critical path analysis

8. Access project results and Experiences

- project review should be conducted to

answer the following fundamental questions

— Did the final product meet or exceed user expectations

— Did the project come in schedule?

— Did the project come within budget?

Exercise

①

Answer:

The project should be considered successful

because, we know that,

A project is considered a success if -

- The system is delivered "on time".
- "within budget".

• The system development had a minimal impact on ongoing business operations

• The resulting information system is acceptable to the customer.

As the completed project fulfills all these criteria it should be considered a success.

Q11/ Answer:

The facts that contributed most to the user dissatisfaction is most likely to be

- poor expectation management

- The project had been in run for almost two years.

- Over time, expectations of the users and managers may have changed

and lead to

- scope creep;

unexpected growth of user

expectations and business requirements

- Feature Creep;

Exam) Chart:

Sl	Task Name	Sept	Oct	Nov	Dec	Jan	Feb	March	April	May	June
	Prob. Analysis										
	Logical Analysis										
4	Technical Analysis										
3	logical design										
5	physical Design										
6	construction & testing										
7	Implementation & Delivery										

April 2021

[6] Answer:

- The Answer would be NO because it will create Feature Scope, which is a part of poor expectation management.

8

An expectations management matrix is a rule-driven tool for helping management understand the dynamics and impact of changing project parameters such as cost, schedule, scope, quality.

A Basic Matrix has 3 rows representing:

- cost
- schedule
- scope and/or quality

And 3 columns correspond to priorities as follows:

Follows:

- Maximize or minimize

↳ The measure of success that is determined to be the most important for a given project.

- constraint

→ 2nd

• Accept

Least Important

Prohibited measure of success	max on min	constraint	Accept
cost		X	
schedule	X		
scope and/or quality			X

8

This would result in rearrangement in

expectation matrix

	max on min	constraint	Accept
cost			X
schedule	X		
scope		X	

10] This will lead to poor expectation management

and in the column, this will result in having 2 knowers (X).

And the schedule must be readjusted on

more money must be allocated to the project.

			X -
	X +		
		X +	

10

Technique used — critical path

11

[10]

The critical path

The sequence of dependent tasks that have the longest sum of most likely durations.

The critical path determines the earliest possible completion date of the project.

Slack Time

The amount of delay that can be tolerated between the starting time and completion time of a task without causing a delay in the completion date of the entire project.

$$A \rightarrow D$$

$$B \rightarrow E$$

$$C \rightarrow D \rightarrow F$$

$$A \rightarrow E \rightarrow G$$

$$H \rightarrow I$$

→ paths will be —

$$A \rightarrow D = 2 + 4 = 6$$

$$B \rightarrow E = 2 + 5 = 7$$

$$C \rightarrow D \rightarrow F = 1 + 4 + 1 = 6$$

$$A \rightarrow E \rightarrow G = 2 + 5 + 6 = 13$$

$$F \rightarrow H = 1 + 4 = 5$$

$$G \rightarrow H \rightarrow I = 6 + 4 + 7 = 17$$

↓
critical path